Agg. Impact

Tester

Aimil Ltd.

Instrumentation & Technologies

Version

2.0

Aimil Ltd.
Instrumentation & Technologies

AIM - 456-1
AGGREGATE IMPACT TESTER

AIM - 456
AGGREGATE IMPACT TESTER
(with Counter)

Operating Instructions

Operating Instructions
Aggregate Impact Tester

Introduction

The equipment is manufactured to meet the essential requirements of IS : 2386 Part IV for determining the Aggregate Impact Value of coarse aggregate. The Aggregate Impact Value gives a relative measure of the resistance of an aggregate to the sudden shock of impact. This machine is furnished with a quick release trigger mechanism to manipulate an impact with a free fall of a 13.75 kg hammer from a height of 380mm ± 5mm. It is supplied complete with one cylindrical cup, one metal measure of 75mm diameter x 50mm depth and a tamping rod.

AIM-456 : This has an additional attachment of an automatic blow counter.

Description

The equipments are illustrated in the General Assembly Drawing. The numbers given against the components of the machine in the description below pertain to this General Assembly Drawing.

The instrument is of rigid construction with a circular base (1) over which the two vertical steel guides (4) are mounted. The guides are connected at the top by a metal plate through which the adjustable stop for release (8) of the hammer (5) and the locking pin (9) operate. The hammer (5) slides up and down through vertical gides and is held by the release claw (6) fixed to the lifting handle (7). The height of the fall of the hammer can be adjusted through 380mm ± 5mm. The cylindrical steel cup (3) can be fixed firmly to the base and is easily removable for emptying. Another cylindrical metal measure (10) of sufficient rigidity is used for the preparation of the test sample which is compacted into measure using a tamping rod (11).

AIM-456 Aggregate Impact Tester with counter is similar to AIM-455 Aggregate Impact Tester described above except that a automatic blow counter (12) is fixed on the top metal plate holding the two guide bars (4) in position. The counter has a lever on the front side and a zero adjustment knob at the rear.

The counter scale is set to the zero reading by rotating the zero adjustment knob anti-clockwise, a striker plate is fixed on the release claw (6) and actuates the lever of the blow counter to record the number of blows.

Setting Up

Mount the apparatus on a plane concrete block of size 600 x 600 x 450mm. The apparatus shall be prevented from rocking by securing it firmly and evenly to the foundation. Three bolts are provided in the base plate for this purpose. Alternatively the apparatus may be supported on a plane metal plate cast into the foundation.

Apply petroleum jelly to the vertical guide columns and lift the hammer with the handle (7), and lock it at the top with the help of the locking pin (9). After having filled the cup with the sample as described in the test procedure below, adjust the height of fall of the hammer through 380 ± 5mm and set the adjustable stop for release in that position. The height of fall is reckoned from the lower surface of the hammer to the upper surface of the sample.

Preparation of the Test Sample

Collect the aggregate passing the 12.5mm IS Sieve and retained on the 10mm IS Sieve as a test sample. Dry it in an oven at 100°C for 4 hours. Fill the metal measure (10) in three layers, tamping each layer with 25 strokes of the rounded end of the tamping rod (11). Strike off the surplus aggregate after compacting the final layer using the tamping rod as a straight edge. Weigh the aggregate in the measure to the nearest gram (weight A). This weight of aggregate shall be used to conduct a duplicate test on the same material.

Test Procedure

Fix the cup on the base firmly and place the whole of the test sample in it and compact it by 25 strokes of the tamping rod.

Raise the hammer unit so that the lower face of the hammer is 380mm above the upper surface of the aggregate in the cup and allow it to fall freely on to the aggregate. Subject the test sample to a total of 15 such blows each being delivered at an interval of not less than one second.

Remove the crushed aggregate from the cup and sieve the whole of it on the 2.36mm IS Sieve until no further significant amount passes in one minute. Weigh the fraction passing the sieve to an accuracy of 0.1 g (weight B). Weigh the fraction retained on the sieve (weight C) and if the total weight (B + C) is less than the initial weight (weight A) by more than one gram, the result should be discarded and a fresh test conducted. Two tests should be conducted.

Calculations

The ratio of the weight of fine particles (fines) formed to the total sample weight in each test shall be expressed as percentage, the result being recorded to the first decimal place.

Aggregate Impact Value = (B / A) x 100%

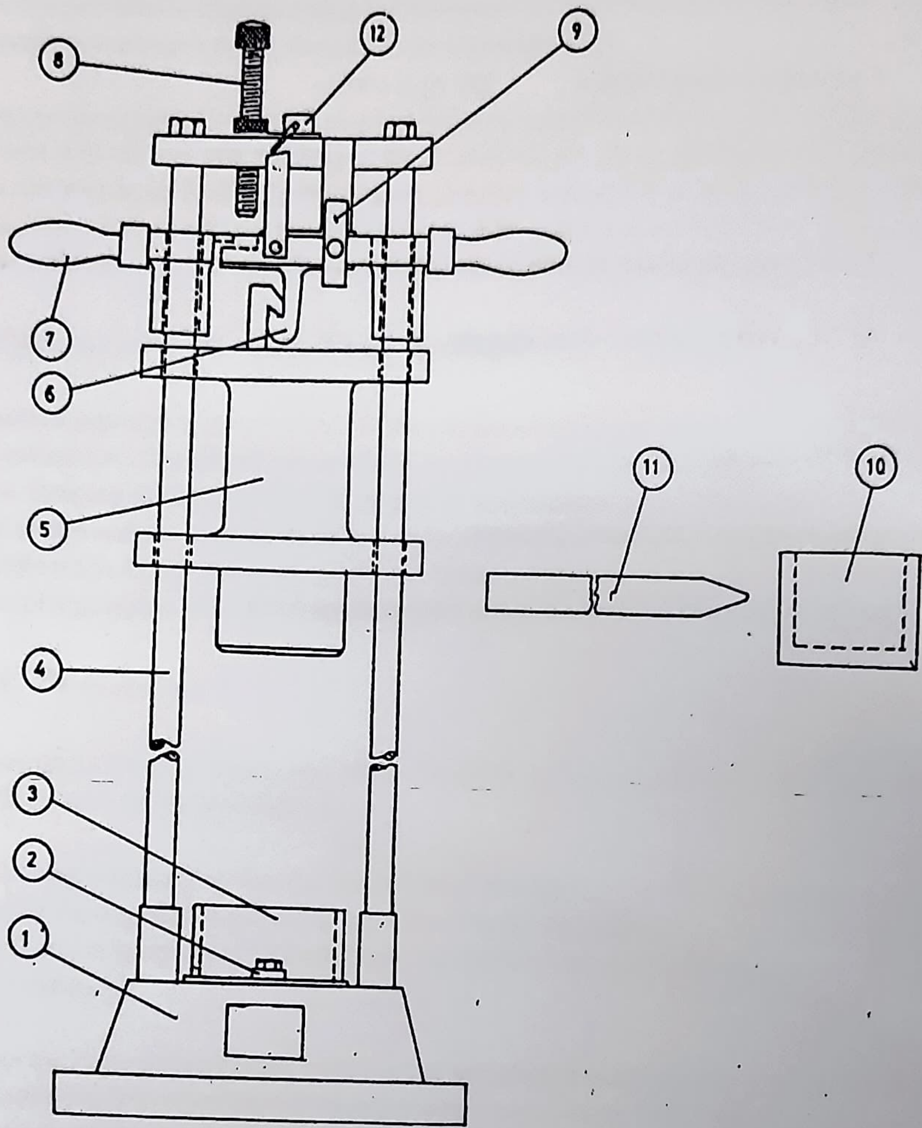
Where

B = Weight of the fraction passing 2.36m IS Sieve

A = Weight of over-dried sample.

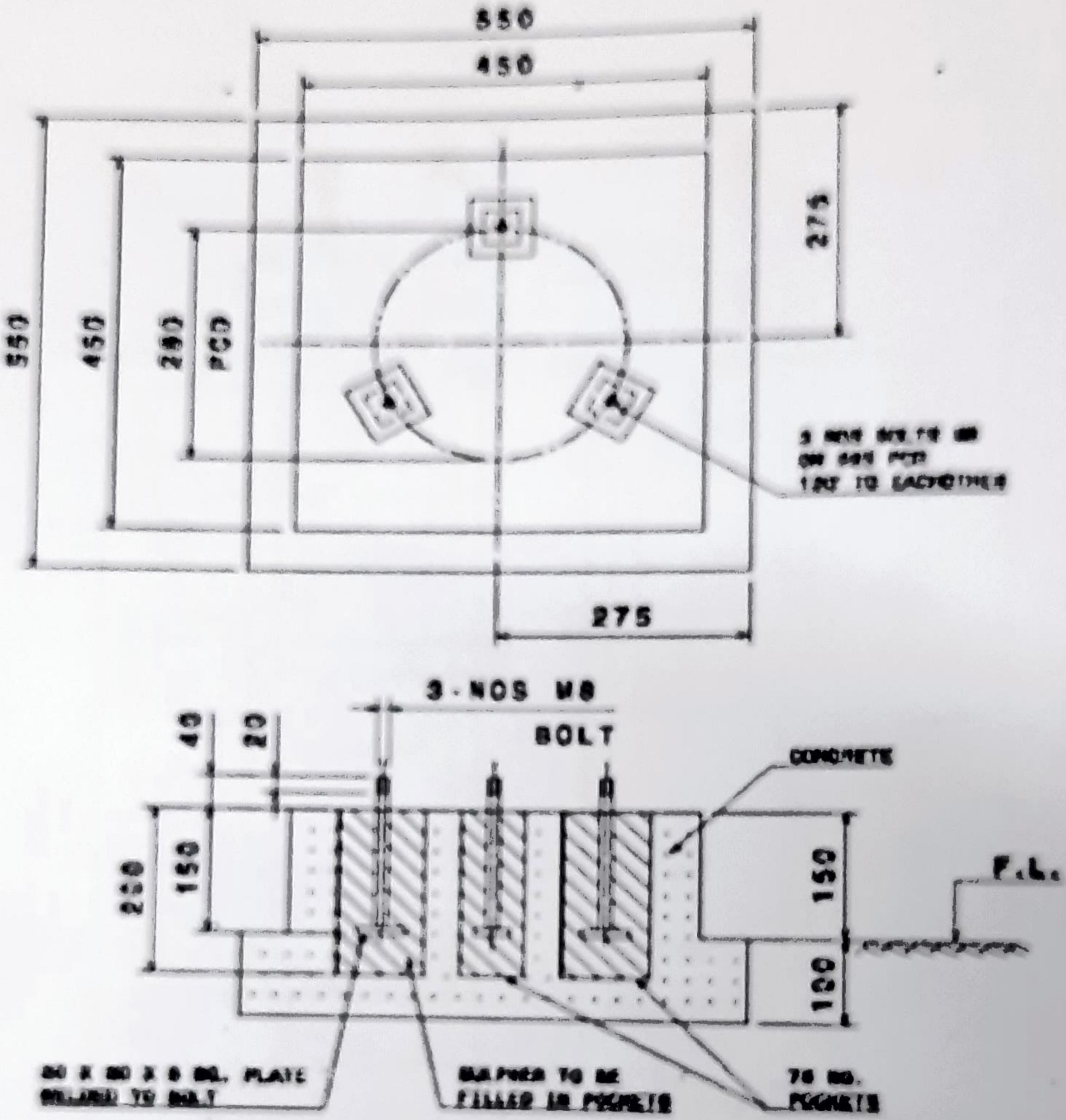
Maintenance

1. Keep all exposed metal parts greased.
2. Keep the guide rods firmly fixed to the base and top plate.



S.N.	DESCRIPTION
1	CIRCULAR BASE
2	CLAMP FOR HOLDING STEEL CUP
3	CYLINDRICAL STEEL CUP
4	TUP GUIDE BARS
5	HAMMER
6	RELEASE CLAW

S.N.	DESCRIPTION
7	LIFTING HANDLES
8	ADJUSTABLE STOP FOR RELEASE
9	LOCKING PIN
10	METAL MEASURE
11	TAMPING ROD
12	BLOW COUNTER



AIM-456

FOUNDATION DRAWING
AGGREGATE IMPACT VALUE TESTER

ALL DIMENSIONS IN MM